Introductory dynamics: cram sheet

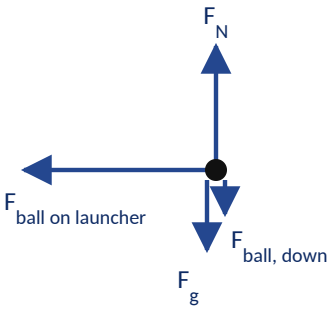
AP Physics 1 · 50 min · 4 problems, 25% each

The four problem types

- **Recoil**: solve for the launcher's acceleration.
- **Thrust and buoyancy**: thrust given, solve for the buoyant force.
- **Free-body diagram on a slope**: drawing only, four forces, no angle needed.
- **Drag linearization**: graph F_D against v^2, K from the slope.

Diagram first, then $a = F_{net}/m$ with the real forces and signs, letters first, then numbers with units.

1. Recoil



a_{ball} from kinematics
→ $F = m_{ball}a_{ball}$ (2nd law)
→ same F back on the launcher (3rd law)
→ $a_{launcher} = F/M$, backward

- Vertical forces balance ($F_N = F_g$ + the ball pressing down); they stay out of the horizontal equation.
- Equal forces, unequal accelerations.
- Match frames: ground speed with ground distance. If the speed is relative to the launcher: $a_{rel} = v^2/(2L) = a_{ball} + a_{launcher}$ and $m_{ball}a_{ball} = M\,a_{launcher}$.

Practice versions assume the ball starts from rest, constant acceleration during the launch, and a freely rolling or sliding launcher, and say which frame each value is in.

2. Thrust and buoyancy



Up is +: $F_b + F_t - mg = ma$
→ $F_b = m(g + a) - F_t$

- No F_N: it has left the bottom.
- a from the motion data, with its sign. Rising while slowing: $a < 0$.

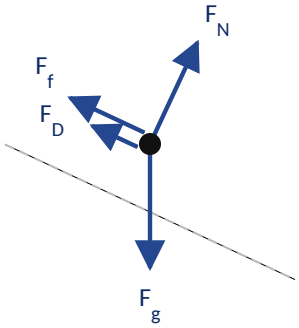
Practice versions assume negligible fluid drag and state the acceleration or the data that give it.

Kinematics (from rest unless stated)

Given	Use	a
v and t	$v = v_0 + at$	$(v - v_0)/t$
v and Δx	$v^2 = v_0^2 + 2a\Delta x$	$(v^2 - v_0^2)/(2\Delta x)$
Δx and t	$\Delta x = v_0t + \frac{1}{2}at^2$	$2\Delta x/t^2$ (from rest)
constant speed	no change in v	0

Conversions: 1000 g = 1 kg · 100 cm = 1 m · 1000 ms = 1 s · km/h ÷ 3.6 = m/s

3. Free-body diagram on a slope (drag and friction count)



- F_g: straight down.
- F_N: perpendicular to the slope, out of the surface.
- F_f: uphill, opposite the sliding.
- F_D: opposite the motion through the air; uphill in still air.

- Exactly four arrows. Components of F_g are not extra forces; drag is the air resistance, not a fifth force.

No angle is needed. Arrow lengths here are not to scale. Practice versions assume still air.

4. Drag linearization

y (vertical)	x (horizontal)	slope
F_D (N)	v^2 (m^2/s^2)	K

- $F_D = K(v^2) + 0$ matches $y = mx + b$.
- Square every speed; best-fit line with a ruler, more than half the grid.
- Slope from two well-separated points on the line, not data points.
- K in $N{\cdot}s^2/m^2 = kg/m$.
- v^2 is speed squared (m^2/s^2), not an acceleration.

Key: F_N normal force · F_g gravity (weight) · F_f kinetic friction · F_D drag · F_t thrust · F_b buoyant force · F_{net} net force (ΣF) · m, M mass of the ball, mass of the launcher · a_{rel} the ball's acceleration relative to the launcher · L the launch length, measured on the launcher